

Flexible Formation Control Using Hausdorff Distance: A Multi-agent Reinforcement Learning Approach

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Abstract—While fixed topology formation control with a centralized controller has been studied for multi-agent systems, it remains challenging to develop robust distributed control policies that can achieve a flexible formation without a global coordinate system. In this paper, we design a fully decentralized displacement-based formation control policy for multi-agent systems, which can achieve any formation after one-time training. In particular, we use a model-free multi-agent reinforcement learning (MARL) approach to obtain such a policy in the centralized training process. The Hausdorff distance is adopted in the reward function for measuring the distance between the current and target topology. The feasibility of our method is verified by both simulation and implementation on omni-directional vehicles.

Index Terms—Multi-agent reinforcement learning (MARL), formation control, Hausdorff distance

I. INTRODUCTION

As a basic task of multi-agent cooperation, the control of multi-agent systems to achieve the desired topology has been widely investigated for their applications in Unmanned Aerial Vehicles (UAV) control [1], [2], Internet of Vehicles [3] and rescue operations [4]. The main challenges for distributed formation control arise from controlling the system based on local and partial information without the aid of a central controller. It introduces additional difficulties when the control flexibility is considered, which requires the formation pattern of agents to be time-varying, such as the rotation formation [5].

Extensive research has been done in the field of formation control using different approaches, such as leader-follower [6], virtual structure [7], potential function-based approach [8] and behavior modeling based control [9]. Those approaches usually work well when a global coordinate system is available and the environment is obstacle-free, but they are not stable in harsh environments. Reinforcement learning (RL) methods have been introduced to overcome the difficulties of non-convex and global observation such as model predictive control (MPC) [10], [11] or fuzzy control [12]. However, a flexible distributed formation control of homogeneous agents, where the ideal formation varies over time, has not been carefully investigated. One challenge is the design of a feasible

reward function to quantify the distance of node-independent topology.

This paper focuses on fully distributed formation control of agents, which gives the optimal control policy based on the displacement of other agents and ideal topology. Compared with previous works, the positions of agents and landmarks under a shared global coordinate system are not required. We propose a new multi-agent reinforcement learning (MARL) approach to learn the position control policy to achieve decentralized control under the full distribution assumption. This policy will enable self-adaptive formation control for mobile robots when central controllers are not available. Our experiment shows that the learned policy can quickly build the target structure according to the given topology. We also conduct our method on a hardware platform using smart cars with omni-directional geometry as agents.¹

The strengths of our solution are as follows: 1) Efficiency: Our strategy can shorten formation time by reducing unnecessary movement. We quantify the difference between the current and target formations by Hausdorff distance and design a reward function accordingly to encourage agents to optimize directly toward the target. 2) Flexibility: The target formation is treated as a part of the observation, so that after delicate one-time training, the agent can switch to any formation conveniently without any re-training.

The paper is organized as follows. In Section 2, related work and current difficulties are reviewed. The problem formulation and theoretical analysis are presented in Section 3. In Section 4, the general approach is introduced. Simulation and experimental results are provided in Section 5. Finally, we draw our conclusion in Section 6.

II. RELATED WORKS

A. Deficiencies of Classical Algorithm

Displacement-based formation control [13] is studied as an optimization problem [14] by researchers over the past decades. In displacement-based formation control, all the agents only have access to their local coordinate systems with

¹The results and codes can be found at <https://github.com/jc-bao/gym-formation>.

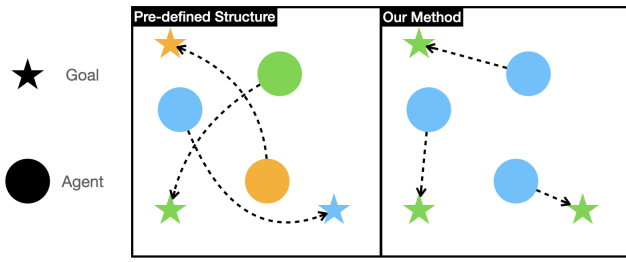


Fig. 1. The circles with different colours represent different agents, and the pentagrams represent landmarks in the target topology. As shown on the left, the agents with a pre-defined policy move an extra distance to get into the target topology compared with the agents with our policy as shown on the right.

a global orientation. The agents must actively control their relative displacement to other agents to achieve the desired formation. Previous studies have proven the convergence in multi-agent systems with general linear and Lipschitz non-linear dynamics [15]. authors proved the local exponential stability for rigid formation systems in [16] and extended it to consider the prescribed orientation in [17]. These approaches work well in obstacle-free environments but need to determine the agent's position in the topology in advance. Fig. 1 shows the difference between the pre-defined policy and our policy.

B. Difficulties in MARL Algorithm

In MARL settings, multiple autonomous agents interact with each other in the same environment, and the algorithm is designed to obtain the sequential decision-making policy that maximizes each agent's reward [18]. MARL has become a popular tool in communication networks [19], social science [20] and vehicle control [21]. MARL is introduced to design the formation control policy by modelling the multi-agent control problem as a partially observed Markov decision process problem [22]. In our formation control task, the cooperative setting with homogeneous agents is adopted, in which agents cooperate to optimize a common reward function.

In MARL formation control tasks, the most commonly used method is leader-follower structure [23] to achieve formation control and obstacle avoidance which will often fail in leader collapse. Decentralized multi-agent formation control via MARL is also studied [24]. Still, the formation is not homogeneous to all agents, which means each agent has a fixed position in the topology identified, leading to unnecessary movement and additional collision risk.

III. PROBLEM FORMULATION

The displacement-based multi-agent formation control problem is considered in both omni-directional [25] and Ackermann-steering gear control [26] environments. Agents achieve the desired topology by actively adjusting the displacement of their neighbours. We assume that each agent can obtain the relative positions of the other agents and the target formation topology in the existence of the orientation of the global coordinate system [13].

Consider a system of N agents, and let $\mathcal{N} = \{1, 2, \dots, N\}$ to be the index set of agents. At each discrete time step t , agent $i \in \mathcal{N}$ needs to compute the control command based on its current observation $o^i(t)$ to minimize the distance between the target and current formation topology.

$$\begin{aligned} \mathcal{T} : \mathcal{S} \times \mathcal{A}_1 \times \dots \times \mathcal{A}_N &\mapsto \mathcal{S} \\ \mathbf{a}(t) &\sim \pi_\theta(\mathbf{a}(t) \mid o(t)) \end{aligned}$$

where θ denotes the parameter of the control policy.

Agent i has no access to every other agent's state and policy in the partially observed settings. The observation vector is composed of four parts: the target topology $o^{\text{tar}}(t)$, the velocity of the agent $o^v(t)$, the relative position of other agents $o^{\text{agent}}(t)$, and the relative position of obstacles $o^{\text{obs}}(t)$.

Therefore, the multi-agent formation control problem can be formulated as a partially observed Markov decision problem (POMDP) [22], and our goal is to optimize the controller parameter θ to get the best reward [27],

$$J_e(\mu_i) = \mathbb{E}_{s \sim p^\mu, a \sim \mu_i} [R_i(s, a)]$$

where i represents the index of the agent, p^μ defines the probability of trajectory $\tau = (s_1, a_1, s_2, a_2, \dots, s_{T-1}, a_{T-1}, s_T)$ under all agents' policy $\mu = \{\mu_1, \mu_2, \dots, \mu_N\}$. The agent i with policy μ_i will get reward R_i at each step.

The reward function R_i is designed according to topology distance, target velocity, and collision times, encouraging agents to converge to the target topology and velocity as fast as possible while avoiding collisions. We will introduce them in detail in the next section.

IV. PROPOSED APPROACH

This section will introduce our approach in two aspects: 1) the multi-goal RL setup and 2) the calculation of topology distance to guide policy search.

A. Multi-Goal RL

Compared to previous works, we consider the formation control problem as a goal-conditioned policy learning problem [28], where each agent receives a target topology $\mathcal{T}$ at the beginning of each episode. The agent observes a state $s(t)$, performs an action $a(t)$ and receives a reward conditioned on the input target formation $r(s(t), \mathcal{T})$ at each time-step t . After T time-steps of interactions with environment, each agent i collects a trajectory $\tau_i = (s_{i,1}, a_{i,1}, s_{i,2}, a_{i,2}, \dots, s_{i,T-1}, a_{i,T-1}, s_{i,T})$.

Specifically, the observation space, the action space, and the reward function are designed as follow:

1) *Observation Space*: The observation at time step t consists of three parts: the target topology $o^{\text{tar}}(t)$, the velocity of the agent $o^v(t)$, and the relative position of other agents $o^{\text{agent}}(t)$. To be more specific,

- $o^{\text{tar}}(t)$ is described as N nodes' positions. In 2-dimensional space, $o^{\text{tar}}(t) = \mathcal{T}(t) \in \mathbb{R}^{2N}$ represents the target topology. Each agent only knows the general topology, but does not know which parts of the graph it should go to.

- $o^v(t)$ is the agent's velocity $\mathbf{v}_i = [v_x, v_y]^T$ according to the global coordinate.
- $o^{\text{agent}}(t)$ indicates the agent's relative position to other agents.

2) *Action Space*: The action space of the formation environment is a continuous control space. The action output of each agent $i, i \in \mathcal{N}$ includes 2 control variable: $\mathbf{a} = [a_x, a_y]^T$, where a_x, a_y represent the accelerate of x-axis and y-axis, respectively.

3) *Reward Design*: Our objective contains three parts: minimizing the distance to the target topology, avoiding collision and gaining the target navigation velocity. The reward function is

$$r_i(t) = r_i^{\text{form}}(t) + r_i^{\text{colli}}(t) + r_i^{\text{vel}}(t)$$

- In particular, the robot gets the maximum reward when the formation matches the target formation. $r_i^{\text{form}}(t) = -d_H(X, \mathcal{T})$, where X denotes the agents' topology, $\mathcal{T}$ denotes the target topology and d_H denotes the distance between them. The detail will be covered in the next part.
- The agent will be penalized by in each collision:

$$r_i^{\text{colli}}(t) = \begin{cases} -1 & \text{if } \|\mathbf{p}_i(t) - \mathbf{p}_j(t)\| < 2D \\ 0 & \text{otherwise.} \end{cases}$$

where $\mathbf{p}_i(t)$ is the position of agent i at time step t under global coordinate system, and D denotes the collision margin.

- The last part is the velocity. Agents are encouraged to navigate at the target velocity as a group:

$$r_i^{\text{vel}}(t) = -\|\mathbf{v}_{\text{tar}} - \mathbf{v}_i\|$$

where $\mathbf{v}_{\text{tar}}$ is the target velocity of the formation under global coordinate system and $\mathbf{v}_i$ is the velocity of agent i under global coordinate system.

We adopt centralized training and decentralized execution from MADDPG [27]. This design allows agents to exploit extra global information to learn a better policy in the formation control scenario.

B. Topology Distance

Given the input (ideal topology) and the output (actual topology), we can calculate the distance between them by Hausdorff distance [29].

The Hausdorff distance denotes the longest distance agents need to move to the ideal topology, which is calculated as

$$d_H(X, Y) = \max \left\{ \sup_{x \in X} d(x, Y), \sup_{y \in Y} d(X, y) \right\}$$

where $d(x, Y)$ and $d(X, y)$ denotes the distance from point set X to point set Y . The Hausdorff distance serves as a non-linear operator to measure the mismatch of two sets. It can effectively guide the policy to search for the optimal solution to fit into the topology by reducing the spatial differences between two sets. The application of Hausdorff distance keeps the homogeneity of agents, which encourages agents to actively search for the

TABLE I
TRAINING PARAMETERS.

Max episode length	Total time steps	Learning rate
30	1×10^6	1×10^{-4}
Epsilon greedy	Target network soft update parameter τ	Buffer size
1×10^{-1}	1×10^{-2}	5×10^5
Batch size	Noise rate for sampling	Discount γ
256	0.25	0.95

optimal solution to get the target topology rather than converge to a simple policy that navigates to a fixed point of the target topology.

V. EXPERIMENT AND RESULTS

In this section, we first describe the experimental setup of our algorithm. Then, we show the training results in our formation simulation environment. Our method achieves 100 cm of cumulative Hausdorff distance in 3-agent formation control task and 140 cm of cumulative Hausdorff distance in 4-agent scenario. Finally, we implement our algorithm on a corresponding hardware platform.

A. Experimental Setup

In our model, the actor and critic networks are 3-layer MLPs with 256 hidden units. ReLU activation function is used in the hidden layer, and Tanh is used in the actor output layer. We rescale the action to $[-1, 1]$ before feeding into the critic network and rescale the output of the actor network to $[-a_{\text{max}}, a_{\text{max}}]$ before execution in the environment, where a_{max} denotes the max acceleration of agents. The training environment is designed upon the OpenAI multi-agent particle environment (MPE) [30]. We train a multi-agent formation control environment on a laptop with i5-8259U CPU. The training process takes 3 hours to converge in 3 agents' environments. The hyper-parameters of the algorithm are summarized in Table I.

B. Training Scenarios

To perform the flexible distributed formation control experiments, we design a 2-dimensional multi-agent formation control environment based on MPE [30], which consists of N agents with continuous space and discrete time. Each agent shares identical action and observation space and maximizes a shared return. In a cooperative flexible formation control scenario, this task contains N agents and no obstacles. At each episode, agents must take physical actions to form a demanded topology given local partial observation and are rewarded based on the similarity between the current topology and the target topology.

C. Simulation Results in MPE

We tested our algorithm on 3 and 4 agents scenarios and compared our Hausdorff metric with near reward and point-to-point reward. The near reward is modified from OpenAI

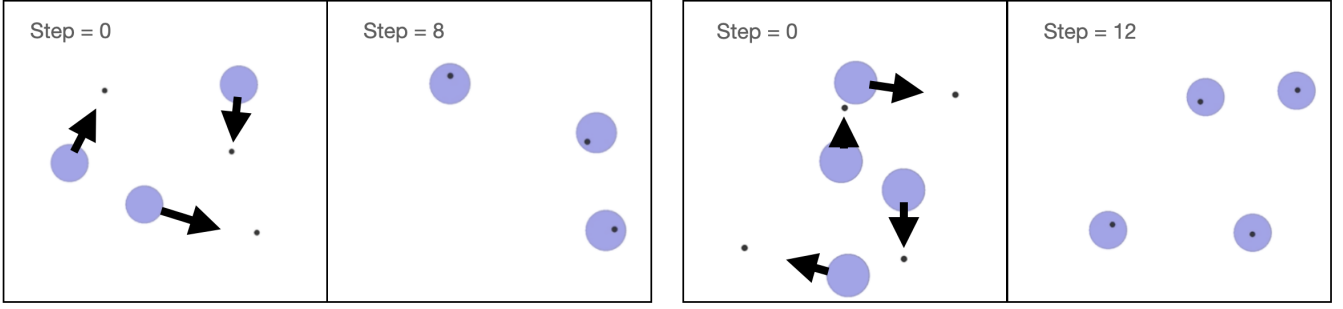


Fig. 2. The purple circles indicate the agent. The black point is the visualization of the ideal topology. At the beginning (step 0), the agents and ideal topology are randomly initialized. Note that the agent cannot directly observe the absolute positions of black dots. **3 Agents Simulation Results.** After 8 time steps, the agents form the target topology. **4 Agents Simulation Results.** Agents need to figure out which part of the topology it should go to minimize formation time. In step 12, the agents get into the ideal topology after several steps of iterations and move together as a team.

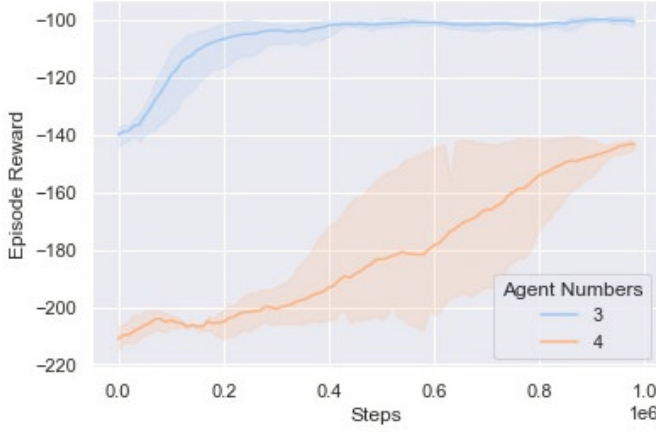


Fig. 3. The curve of the episode reward function (3-agent case and 4-agent case). Due to the longer time horizon and larger distance to the target topology, the 4 agents' reward is lower than 3 agents' case.

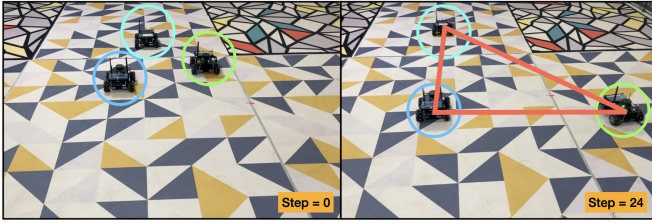


Fig. 4. **Hardware implementation results.** We put three omni-directional vehicles with Mecanum wheels in the middle of the room and control them with our tuned policy. At step 0, the agents were initialized randomly, and the ideal formation is generalized. At step 24, the target triangle topology is formed, and agents move together as a group.

MPE simple spread environment based on the distance of agent to each landmark [30]. Our implementation synchronizes landmarks and agents' centers before calculating the distance. The point-to-point reward is the sum of each agent's distance to the pre-determined landmark.

Table II shows the evaluation results of different reward functions. The Hausdorff distance has lower collision times and needs fewer steps to converge. We have observed that

TABLE II
MEDIAN AND STANDARD DEVIATION ON AGENT'S CONFLICT TIMES AND FORMATION STEPS WITH DIFFERENT REWARD FUNCTIONS.

Reward Type	Near	Point-to-point	Hausdorff (ours)
3-agent formation steps	4.25±1.40	3.89±1.48	3.74±1.37
4-agent formation steps	N/A	6.91±6.54	4.76±1.38
3-agent collision times	0.040±0.154	0.044±0.196	0.016±0.103
4-agent collision times	N/A	0.057±0.198	0.057± 0.178

agents learned a self-regulating consensus strategy that enables faster convergence and fewer conflicts. Agents learn to balance the teams' move distance rather than greedily heading to the nearest point. In the 4-agent scenario, the agents acquire the policy of changing the moving direction in the mid-stage when two agents move to the same point. We contribute this emergent behavior to the Hausdorff distance metric, which only counts for the effective movement that contributes to the whole team.

Fig. 2 shows how agents decide which part of formation they should fit into and move to form the target topology with the shortest path. Our open-source formation control environment and testing results can be found at <https://github.com/jc-bao/gym-formation>.

D. Hardware Implementation

The challenge in adapting policy learned in simulation to the real world is the uncertainty of parameters between real and simulated domains. We successfully implemented our algorithm in a Mecanum wheel multi-vehicle system by randomizing physical dynamic parameters [31], including the mass of each agent, the friction of the ground and the observation noise. In the hardware implementation, the control rate is set to be 10Hz, the region of speed is constrained to 0.5m/s,

and the testing ground is around $4m \times 4m$. The observation is given by the OptiTrack system.² The target topology is sent by the Host computer, which is also generalized randomly to test the flexibility of the control algorithm. Demos can be found at <https://tinyurl.com/FlexibleFormation>.

VI. CONCLUSION

This paper developed an efficient and flexible multi-agent formation control framework to achieve changeable topology. Specifically, we introduced Hausdorff distance into the MARL training process and treated the target topology as a part of the observation. The agents can reach a consensus and move to the nearest goal point avoiding extra distance caused by the pre-defined policy. Our method can also easily switch the target formation topology without re-training. We also validated our method in both simulation and hardware implementation.

Future work will explore harsh environments with moving obstacles, observation-constrained scenarios, and adjustable formation control with only raw sensor data. Moreover, combining the global motion controller and local formation controller using hierarchical RL is also a promising approach to solving multi-agent formation control and navigation problems.

VII. ACKNOWLEDGEMENT

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REFERENCES

- [1] B. D. O. Anderson, B. Fidan, C. Yu, and D. Walle, "UAV formation control: Theory and application," in *Recent Advances Learn. Control*, 2008, pp. 15–33.
- [2] Y. Yan, X. Li, X. Qiu, J. Qiu, J. Wang, Y. Wang, and Y. Shen, "Relative distributed formation and obstacle avoidance with multi-agent reinforcement learning," in *Proc. IEEE Int. Conf. Robot. Autom.*, Philadelphia, PA, United States, May 2022, pp. 1–6.
- [3] H. Raza and P. Ioannou, "Vehicle following control design for automated highway systems," *IEEE Control Syst. Mag.*, vol. 16, no. 6, pp. 43–60, Aug. 1996.
- [4] J. Saez-Pons, L. Alboul, J. Penders, and L. Nomdedeu, "Multi-robot team formation control in the guardians project," *Ind. Robot: An Int. J.*, Jun. 2010.
- [5] X. Dong, B. Yu, Z. Shi, and Y. Zhong, "Time-varying formation control for unmanned aerial vehicles: Theories and applications," *IEEE Trans. Control Syst. Technol.*, vol. 23, no. 1, pp. 340–348, Apr. 2014.
- [6] L. Consolini, F. Morbidi, D. Prattichizzo, and M. Tosques, "Leader-follower formation control of nonholonomic mobile robots with input constraints," *Automatica*, vol. 44, no. 5, pp. 1343–1349, May 2008.
- [7] M. A. Lewis and K.-H. Tan, "High precision formation control of mobile robots using virtual structures," *Auton. Robots*, vol. 4, no. 4, pp. 387–403, Oct. 1997.
- [8] D. Xue, J. Yao, G. Chen, and Y.-L. Yu, "Formation control of networked multi-agent systems," *IET Control Theory and Appl.*, vol. 4, no. 10, pp. 2168–2176, Jan. 2010.
- [9] T. Balch and R. C. Arkin, "Behavior-based formation control for multirobot teams," *IEEE Trans. Robot. Autom.*, vol. 14, no. 6, pp. 926–939, Dec. 1998.
- [10] X. Li, K. Ma, J. Wang, and Y. Shen, "An integrated design of cooperative localization and motion control," in *Proc. IEEE Int. Conf. Commun. in China*, Changchun, China, Aug. 2019, pp. 1–5.
- [11] L. Dai, Q. Cao, Y. Xia, and Y. Gao, "Distributed mpc for formation of multi-agent systems with collision avoidance and obstacle avoidance," *J. Franklin Inst.*, vol. 354, no. 4, pp. 2068–2085, Mar. 2017.
- [12] J. E. Naranjo, C. Gonzalez, R. Garcia, and T. D. Pedro, "Lane-change fuzzy control in autonomous vehicles for the overtaking maneuver," *IEEE Intell. Transp. Syst. Mag.*, vol. 9, no. 3, pp. 438–450, Sep. 2008.
- [13] K.-K. Oh, M.-C. Park, and H.-S. Ahn, "A survey of multi-agent formation control," *Automatica*, vol. 53, pp. 424–440, Mar. 2015.
- [14] Y. Cao, W. Yu, W. Ren, and G. Chen, "An overview of recent progress in the study of distributed multi-agent coordination," *IEEE Trans. Ind. Informat.*, vol. 9, no. 1, pp. 427–438, Feb. 2012.
- [15] Z. Li, W. Ren, X. Liu, and M. Fu, "Consensus of multi-agent systems with general linear and lipschitz nonlinear dynamics using distributed adaptive protocols," *IEEE Trans. Autom. Control*, vol. 58, Sep. 2011.
- [16] Z. Sun, S. Mou, B. D. Anderson, and M. Cao, "Exponential stability for formation control systems with generalized controllers: A unified approach," *Syst. Control Lett.*, vol. 93, pp. 50–57, Jul. 2016.
- [17] Z. Sun, M.-C. Park, B. D. Anderson, and H.-S. Ahn, "Distributed stabilization control of rigid formations with prescribed orientation," *Automatica*, vol. 78, pp. 250–257, Apr. 2017.
- [18] K. Zhang, Z. Yang, and T. Başar, "Multi-agent reinforcement learning: A selective overview of theories and algorithms," *Handbook of Reinforcement Learn. and Control*, pp. 321–384, 2021.
- [19] J. Cortes, S. Martinez, T. Karatas, and F. Bullo, "Coverage control for mobile sensing networks," *IEEE Trans. Robot. Autom.*, vol. 20, no. 2, pp. 243–255, Apr. 2004.
- [20] C. Castelfranchi, "The theory of social functions: Challenges for computational social science and multi-agent learning," *Cogn. Syst. Res.*, vol. 2, no. 1, pp. 5–38, Apr. 2001.
- [21] G. Wen, C. P. Chen, J. Feng, and N. Zhou, "Optimized multi-agent formation control based on an identifier-actor-critic reinforcement learning algorithm," *IEEE Trans. Fuzzy Syst.*, vol. 26, no. 5, pp. 2719–2731, Dec. 2017.
- [22] W. S. Lovejoy, "A survey of algorithmic methods for partially observed markov decision processes," *Ann. Oper. Res.*, vol. 28, no. 1, pp. 47–65, Dec. 1991.
- [23] G. Wen, C. L. P. Chen, J. Feng, and N. Zhou, "Optimized multi-agent formation control based on an identifier-actor-critic reinforcement learning algorithm," *IEEE Trans. Fuzzy Syst.*, vol. 26, no. 5, pp. 2719–2731, Dec. 2018.
- [24] A. Gutpa and R. Nallanthighal, "Decentralized multi-agent formation control via deep reinforcement learning," in *Int. Conf. Agents Artif. Intell.*, Vienna, Austria, Feb. 2021, pp. 289–295.
- [25] G. Indiveri, "Swedish wheeled omnidirectional mobile robots: Kinematics analysis and control," *IEEE Trans. Robot.*, vol. 25, no. 1, pp. 164–171, Feb. 2009.
- [26] W. C. Mitchell, A. Staniforth, and I. Scott, "Analysis of ackermann steering geometry," SAE Technical Paper, Tech. Rep., 2006.
- [27] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, Dec. 2017, pp. 6382–6393.
- [28] T. Schaul, D. Horgan, K. Gregor, and D. Silver, "Universal value function approximators," in *Int. Conf. Mach. Learn.*, Lille, France, Jul. 2015, pp. 1312–1320.
- [29] D. Huttenlocher, G. Klanderman, and W. Rucklidge, "Comparing images using the hausdorff distance," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 15, no. 9, pp. 850–863, Sep. 1993.
- [30] I. Mordatch and P. Abbeel, "Emergence of grounded compositional language in multi-agent populations," in *Assoc. Advancement Artif. Intell.*, New Orleans, Louisiana, USA, 2018, pp. 1495–1502.
- [31] X. B. Peng, M. Andrychowicz, W. Zaremba, and P. Abbeel, "Sim-to-real transfer of robotic control with dynamics randomization," in *Proc. IEEE Int. Conf. Robot. Autom.*, Brisbane, Australia, May 2018, pp. 3803–3810.

²<https://www.optitrack.com>.